

BNU Computer Networks Project, Packet Tracer Development Rules

Purpose: To ensure all departmental topologies can be merged into a single Packet Tracer file without conflicts.

1. DEVICE OWNERSHIP

Muhammad Arez (Integration Lead)

- R-01 Router
- CS-01 Core Switch
- SW-SRV
- SRV-01 • SRV-02 • SRV-03
- SRV-04
- ISP Connection
- VLAN Configuration
- Router-on-a-Stick
- DHCP Relay
- NAT/PAT
- DNS Configuration
- Final Integration

Other Members: Build ONLY assigned departmental topologies.

Do NOT create the core:

- R-01
- CS-01
- ISP
- Server Farm

1. NAMING CONVENTION

Admission: SW-ADM ADM-PC01 to ADM-PC12 ADM-LT01 ADM-SRV01

Exam: SW-EXAM EXAM-PC01 to EXAM-PC15 EXAM-SRV01 EXAM-SRV02

Admin: SW-ADMIN ADMIN-PC01 to ADMIN-PC15 ADMIN-SRV01

Finance: SW-FIN FIN-PC01 to FIN-PC13

Marketing: SW-MKT MKT-PC01 to MKT-PC08 MKT-LT01 to MKT-LT04 AP-MKT

Do NOT use: PC0 PC1 Switch0 Switch1 Server0 warna we'll have pinging and distinguishing problems between each of your endpoint devices after integration

1. VLAN ASSIGNMENTS

Admission = VLAN 10 Exam = VLAN 20 Admin = VLAN 40 Finance = VLAN 50 Marketing = VLAN 60 Server Farm = VLAN 70

This info is also present in our doc report, do not choose your own vlan, use these strictly

Do NOT create new VLAN numbers.

Do NOT rename VLANs.

1. DEVICE REQUIREMENTS Admission:

- SW-ADM
- 12 PCs
- 1 Laptop
- 1 Server Exam:
- SW-EXAM
- 15 PCs
- 2 Servers Admin:
- SW-ADMIN
- 15 PCs
- 1 Server Finance:
- SW-FIN
- 13 PCs Marketing:
- SW-MKT
- 8 PCs
- 4 Laptops
- 1 Access Point

1. CONFIGURATION RULES Department

Members:

- Set correct hostnames.
- Place devices neatly.
- Connect devices to department switch.
- Assign static IPs only where required by report.
- Leave DHCP devices set to DHCP.

Do NOT configure:

- Router-on-a-Stick
- NAT
- DHCP Relay
- DNS
- Core VLAN Trunks
- ISP Connectivity

1. INTEGRATION PROCESS

Each member submits their completed department topology.

I will:

- Import department topologies.
- Connect each department to CS-01.
- Configure VLAN trunks.
- Configure router subinterfaces.
- Configure DHCP Relay.
- Configure NAT/PAT.
- Configure DNS.
- Perform final testing.

OTHER WISE IF EACH OF YOU CREATE YOUR OWN BACKBONE, THERE ARE INTEGRATION PROBLEMS SO FOLLOW THE INSTRUCTIONS EXACTLY

1. FINAL TESTS

Every department must successfully:

Obtain DHCP address Ping default gateway Ping SRV-01 Resolve DNS names Access LMS Server Reach other VLANs Reach Internet through NAT

Failure to follow naming, VLAN, or device rules may cause integration delays and configuration conflicts.

YOU CAN POST THIS LITTLE GUIDE TO YOUR CHATBOT OF CHOICE AND HAVE THEM FEED THE RULES TO YOU SO YOU DONT MISS SOMETHING OR MESS UP